



# The Robots' Guide To World Domination

Going by the effort being put in to make robots smarter, we would be seeing 'intelligent' robots in no time. This article points to some latest technologies that are being developed which could take over this world

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**O**nly five countries in the world actually voted in favour of banning autonomous (a.k.a smart) killer robots at a recent UN convention. This could mean that there are a lot of countries that are already building their own autonomous robots or are in favour of building them soon. In fact, Russia announced that autonomous robots that will gun down trespassers, no questions asked, would soon man their ballistic missile bases.

Of course it is not all about battles and war. Japan's Prime Minister Shinzo Abe was reported by *Jiji Press* as saying, "In 2020, I would like to gather all of the world's robots and aim to hold an Olympics where they compete in technical skill." It would

be an understatement to say that the future will have a lot of autonomous robots. They are already here, being designed and manufactured around the world. And perhaps, they have already left the industry shop-floor and are on their way to your living room.

## Step one to world domination: infiltrate homes

ABI Research says 1.8 million home automation systems will ship this year, and it will increase to 12 million in the next five years. What's more, seemingly unobtrusive electronics like small domestic-lighting bulbs, home heating and cooling, smoke sensors and electronic door-locks would be collecting usage data which the other machines could also make use of.

Companies are aggregating previously-available technologies and putting them into small devices that are

affordable and can be used in domestic scenarios to make life easier at home. Cisco has brought out its Control4 technology which pairs up with Zigbee for home automation. Verizon has its own automation service that is powered by Motorola's technology that it got through its acquisition of 4Home.

While useless individually, big data analytics conducted on accumulated data from the bots in your home lets outside machines and their users gain considerable insight on the behaviour of the people living in your home. These machines can understand when you come home, and also whether you go for a bath immediately or you have food or exercise first. Over a long term, it can understand whether you have other people at home, or if it is time to 'ask' an autonomous vacuum cleaner to clean up your place. It can even determine the best time to break into your house while you are sleeping!

US-based Droplet Robotics has introduced a data-analytics powered system that can identify the soil and plant types to precisely disperse required amount of water for an optimal growth of each plant. It also provides the user with analytics data like how much water goes to each type of plant, tree or lawn. They use cloud computing, connected services and machine-to-machine communication (M2M) to analyse situations and identify events. They also use the information to make intelligent decisions that are then implemented through robots that they control.

Remember Robovac's? These are those intelligent and autonomous cleaning systems of yesteryear that have the capability to analyse their surroundings and use the data to implement the most efficient cleaning process. Things have changed a lot since the introduction of these robots though.

The latest Robovac's like the Roomba 800 series from iRobot have advanced their robots' 'smartness factor' by enabling them to make the most of a whole lot of sensors that come with the bot. The newer robots can detect cliffs and steps before they fall down, know how to fol-